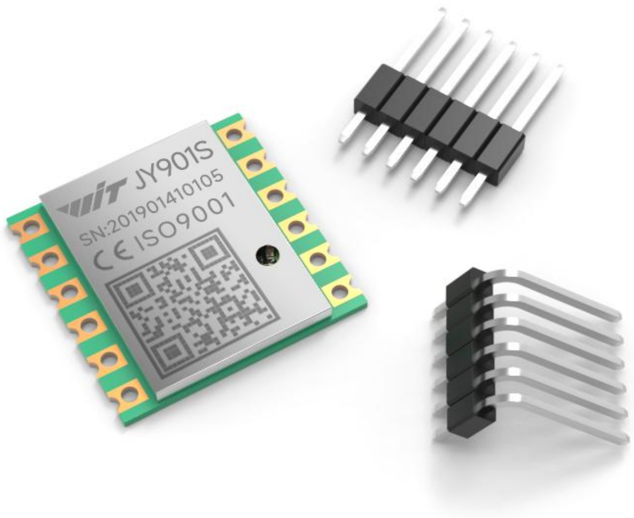
WT901 Data Sheet & User Manual



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1. Product Introduction

1.1. Product Overview

- This product is a high-performance 3D motion attitude measurement system based on MEMS technology. It includes motion sensors such as a three-axis gyroscope , a three-axis accelerometer , and a three-axis electronic compass . By integrating various high-performance sensors and utilizing a proprietary core attitude dynamics algorithm engine, combined with a high-dynamic Kalman filter fusion algorithm, it provides customers with high-precision, high-dynamic, and real-time compensation of three-axis attitude angles. Flexible configuration of various data types allows for diverse application scenarios.

- The leading sensor fusion algorithm based on Kalman filtering principle and with independent intellectual property rights can provide real-time data with an update rate of up to 200Hz, thus meeting various high-precision application requirements and achieving accurate motion capture and posture estimation.

- We have the domestically leading high-precision turntable equipment and instruments, and our products integrate independently developed high-precision calibration and calibration algorithms to improve the measurement accuracy of the products.

- At the same time, we provide users with various host computers, operating instructions, development manuals, and development codes required to minimize the R&D time for various needs.

1.2. Features

- The module integrates high-precision gyroscopes, accelerometers, and geomagnetic field sensors, and uses a high-performance microprocessor and advanced dynamics solver and Kalman dynamic filtering algorithms to quickly calculate the module's current real-time motion posture.

- The use of advanced digital filtering technology can effectively reduce measurement noise and improve measurement accuracy.

- The module integrates an attitude solver, which, combined with the dynamic Kalman filter algorithm, can accurately output the current attitude of the module in a dynamic environment. The attitude measurement accuracy is 0.2°, the stability is extremely high, and the performance is even better than some professional inclinometers.

- The Z-axis heading angle incorporates geomagnetic sensor filtering and fusion, resolving the cumulative error caused by gyroscope integral drift in the 6-axis algorithm. This ensures long-term stable heading angle output.

Note: Due to magnetic field detection, calibration is required before use. During use, the device must be kept at least 20 cm away from magnetic interference areas, electronic devices, magnets, speakers, and other hard magnetic objects.

- The module has its own voltage stabilization circuit, the operating voltage is 3.3~5V, and the pin level is compatible with 3.3V/5V embedded systems, making it easy to connect.
- Supports both serial and IIC digital interfaces, allowing users to easily choose the optimal connection method. The serial port rate is adjustable from 4800bps to 230400bps, and the IIC interface supports full-speed 400K.
- The maximum data output rate is 200Hz. The output content can be selected arbitrarily, and the output rate is adjustable from 0.2 to 200Hz.
- 4 expansion ports are reserved and can be configured as analog input, digital input, digital output and other functions.
- The stamp hole is gold-plated and can be embedded in the user's PCB board. **Note: If you need to add a base plate or embed it on other PCB boards, do not run wires under the geomagnetic chip to avoid interfering with the magnetometer.**
- 4-layer PCB technology, thinner, smaller and more reliable.
- Support low power instruction wake-up and pin wake-up. Sleep power consumption 15ua-35ua

2. Parameter indicators

2.1. Accelerometer parameters

parameter	condition	Typical values
Range		±16g
Resolution	±16g	0.0005(g/LSB)
RMS noise	Bandwidth = 100 Hz	0.75~1mg-rms
Static zero drift	Horizontal placement	±20~40mg
Temperature Drift	-40°C ~ +85°C	±0.15mg/℃
bandwidth		5~256Hz

2.2. Gyroscope parameters

parameter	condition	Typical values
Range		±2000°/s
Resolution	±2000°/s	0.061(°/s)/(LSB)
RMS noise	Bandwidth = 100 Hz	0.028~0.07(°/s)-rms
Static zero drift	Horizontal placement	±0.5~1°/s
Temperature Drift	-40°C ~ +85°C	±0.005~0.015 (°/s)/℃
bandwidth		5~256Hz
Bias stability	10 smoothing (1σ)	3.19°/h
Bias instability	Allan variance	2.24°/h

2.3. Magnetometer parameters

parameter	condition	Typical values
Range		±2Gauss
Resolution	±2Gauss	6.67nT/LSB

2.4. Pitch and roll angle parameters

parameter	condition	Typical values
Range		X:±180°
		Y:±90°
Inclination accuracy		0.2°
Resolution	Horizontal placement	0.0055°
Temperature Drift	-40°C ~ +85°C	±0.5~1°

2.5. Heading angle parameters

parameter	condition	Typical values
Range		Z:±180°
Heading accuracy	9-axis algorithm, magnetic field calibration, dynamic/static	1° (without magnetic field interference) 【1】
	6-axis algorithm, static	0.5° (Dynamic integral cumulative error exists) 【2】
Resolution	Horizontal placement	0.0055°

Note:

【1】Please calibrate the magnetic field in the test environment before use to ensure that the sensor is familiar with the magnetic field in the environment. Please stay away from magnetic interference during calibration.

【2】In some vibration environments, there will be cumulative errors. The specific error cannot be estimated and will be determined based on actual testing.

2.6. Module parameters

2.6.1. Basic parameters

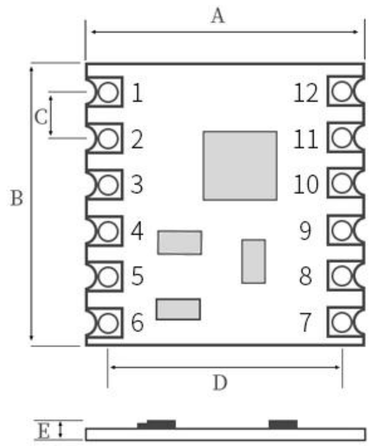
parameter	condition		Minimum	default	Maximum
Communication interface	UART		4800bps	9600bps	230400bps
	Hardware I2C				400K
	Simulating I2C				100K

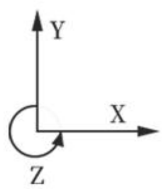
Output content	chip time, 3-axis acceleration, 3-axis angular velocity, 3-axis magnetic field, 3-axis angle, quaternion, port status				
Output rate			0.2Hz	10Hz	200Hz
Startup time					1000ms
Operating temperature			-40℃		85℃
Storage temperature			-40 ℃		100 ℃
impact resistant					20000g

2.6.2. Electrical parameters

parameter	condition	Minimum	default	Maximum
Supply voltage		3.3V	5V	5V
Working current	Working (5V)		11.5mA	
	Sleep (5V)		10.05uA	

3. Product size





Units:mm

	Size
A	15.24
B	15.24
C	2.54
D	12.7
E	2

Stamp hole diameter: 0.9mm

Hole spacing: 2.54mm

4. Pin definition and connection

4.1. Pin Description

Note: Ultrasonic welding, cleaning, cutting, etc. are prohibited on the gyroscope sensor.

Pin number	Pin Name	Pin Function
1	D0	Analog input, digital input and output, pull high to wake up
2	VCC	Power supply 3.3~5V
3	RX	Serial data input
4	TX	Serial data output
5	GND	Ground wire
6	D1	Analog input, digital input and output, setting angle reference
7	D3	Analog input, digital input and output, angle alarm output, X-angle alarm output
8	GND	Ground wire
9	SDA	I2C data line, Y-angle alarm output
10	SCL	I2C clock line, Y+ angle alarm output
11	VCC	Power supply 3.3~5V
12	D2	Analog input, digital input and output, hardware reset, X+ angle alarm output

4.2. Connecting the WT901 module to the PC software

4.2.1. PC software download

1. CH340 driver download link:

https://drive.google.com/file/d/1JidopB42R9EsCzMAYC3Ya9eJ8JbHapRF/view?usp=drive_link

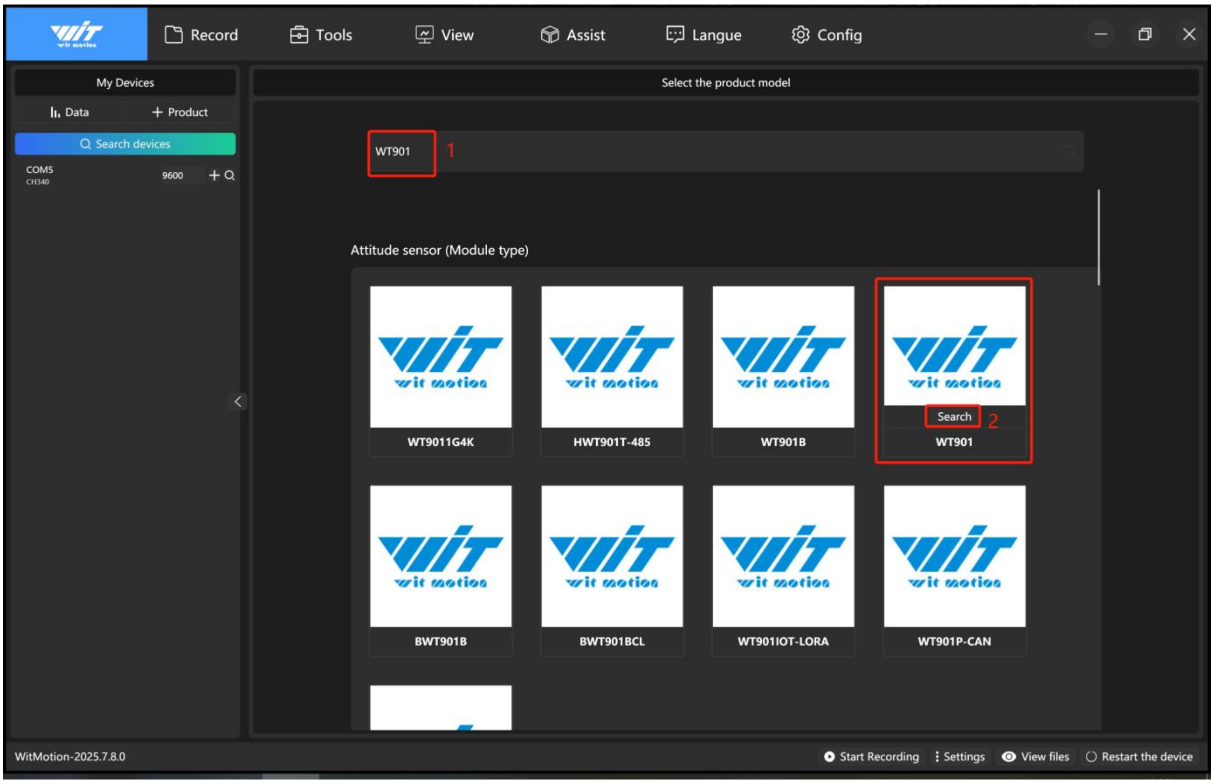
2. Witmotion software download link:
https://drive.google.com/file/d/10xysnkuyUwi3AK_t3965SLr5Yt6YKE-u/view?usp=drive_link

4.2.2. PC software connection method

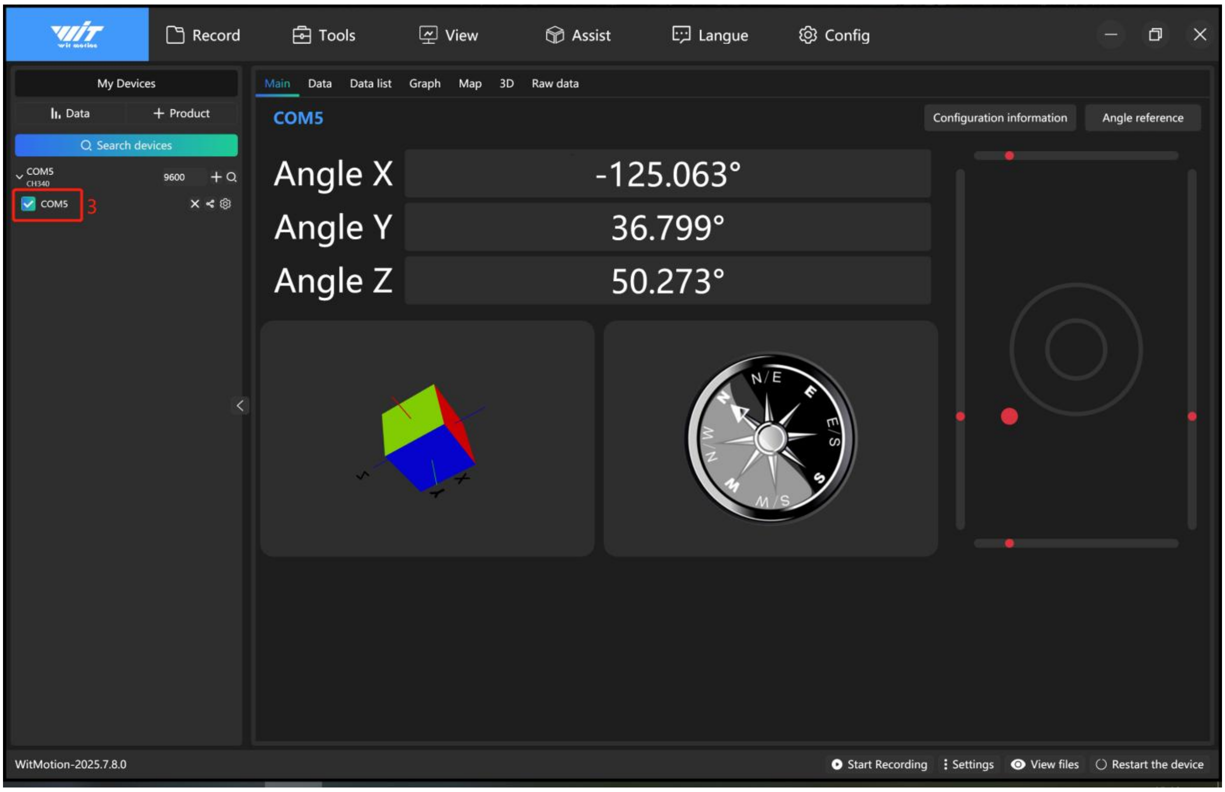
1. Open Witmotion software:

①Select model WT901;

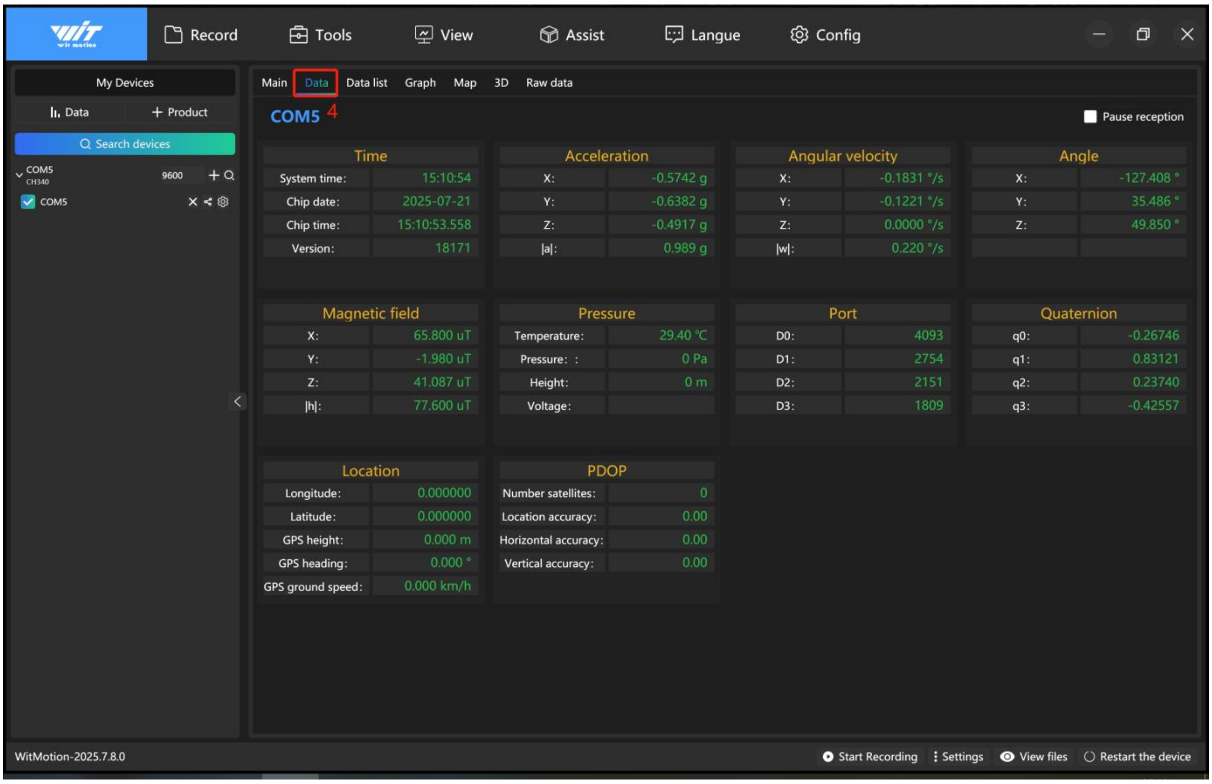
②Click Search;



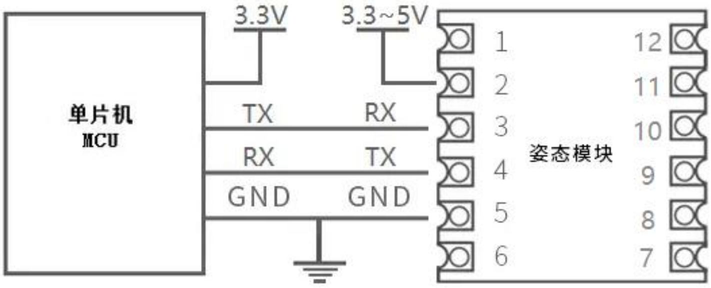
③ Check the serial port number;



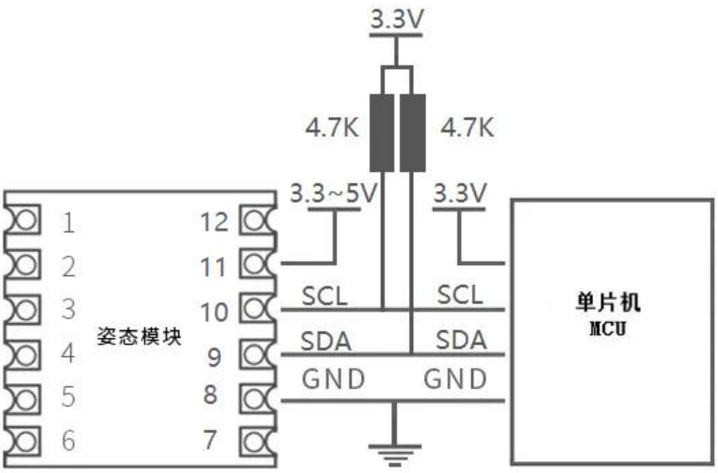
④Switch data interface;



4.3. WT901 module UART connection with MCU



4.4. WT901 module I2C connection with MCU



5. Product Packaging

5.1. Sample packaging

5.2. Bulk packaging